- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

9 In this question, the vectors **i** and **j** are directed east and north respectively.

The velocity $\mathbf{v}\,\text{m s}^{-1}$ of a particle at time t s is given by $\mathbf{v} = kt^2\mathbf{i} + 6t\mathbf{j}$, where k is a positive constant. The magnitude of the acceleration when $t = 2$ is $10\,\text{m s}^{-2}$.

(a) Calculate the value of k. **[4]**

The particle is at the origin when $t = 0$.

(b) Determine an expression for the position vector of the particle at time t. **[2]**

(c) Determine the time when the particle is directly north-east of the origin. **[2]**